

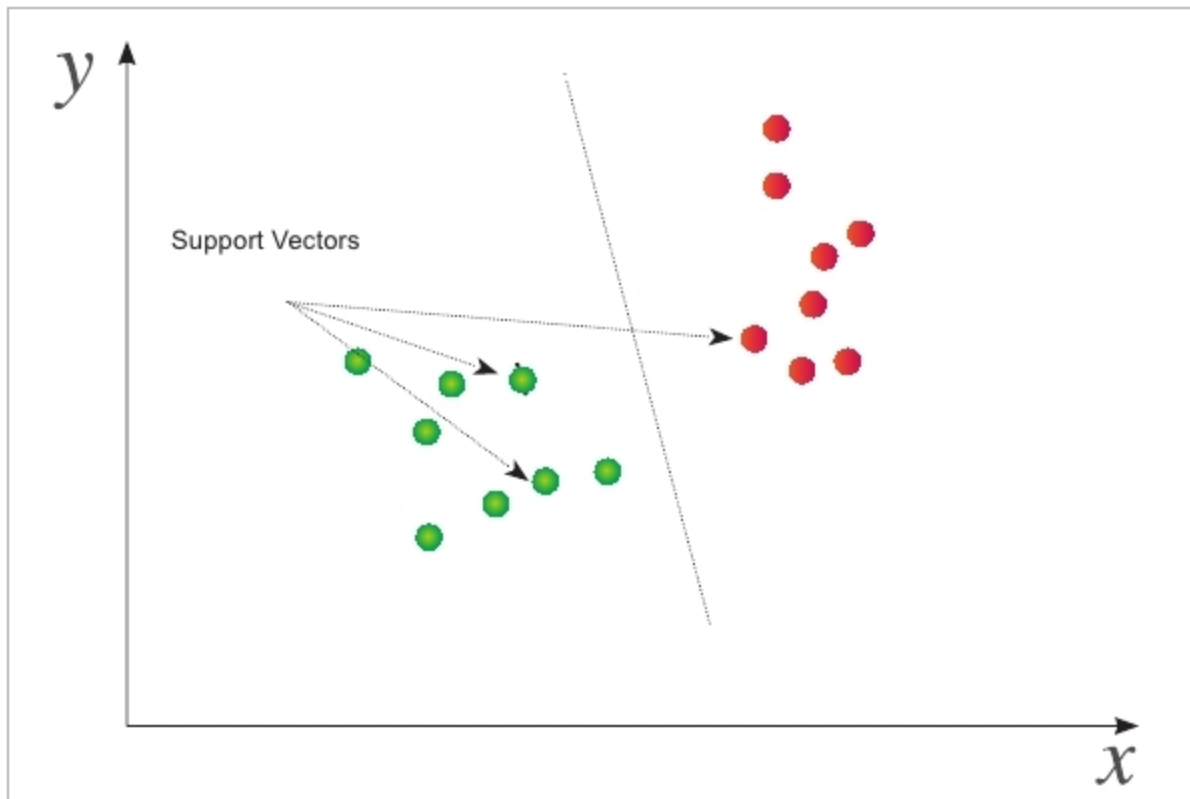


Figure1: Support Vector Machine

Supervised learning, in the context of AI and ML, is a system in which both the input and the desired output data are provided. For example, if you plot millions of sample points in a graph, you can fit a line to approximate a function. Supervised learning systems are mostly associated with retrieval-based AI but they may also be capable of using a generative learning model.

Unsupervised learning is the training of an AI algorithm using information that is neither classified nor labelled and allowing the algorithm to act on that information without guidance. So, when Facebook wants to find out the exact geographical locations where people do not have access to the Internet, all it needs to do is take 350TB of satellite imagery and run it through an AI-enhanced image-recognition engine. After processing 14.6 billion pictures, Facebook will be able to identify more than 2 billion disconnected people across 20 countries.

Artificial intelligence is applied to GIS in areas such as image classification, prediction and segmentation.

Image classification using Support Vector Machine

When you look at a satellite image, it's not always easy to know whether you are looking at trees or grass, at roads or

at buildings. So imagine how difficult it would be for a computer to differentiate between these.

Support Vector Machine (SVM) is a machine learning technique that takes classified data and looks at the extremes. It is a discriminative classifier formally defined by drawing a decision boundary line based on the data called a hyperplane. The data points that the hyperplane margin pushes up against are the support vectors.

Support vectors are important because they are the data points closest to the opposing classes and all other training points can be ignored in the model. When you feed SVM the training data samples of trees and grass, it builds the model, generating a decision boundary of its own. But the results of this supervised classification aren't perfect and algorithms still have a lot more learning to do. They still need to work on features like roads, wetlands and buildings. As algorithms get more training data, they will eventually improve the classification.

Image prediction using Empirical Bayesian Kriging (EBK)

Geostatistics is the generic name for a family of techniques that is used for the mapping of surfaces from limited sample data and the estimation of values

at unsampled locations. Geostatistical estimation is a two-stage process:

- i. Studying the gathered data to establish the predictability of values from place to place (locations) in the study area, which results in a graph known as a semi-variogram.
- ii. The process of estimating values at those locations that have not been sampled is known as kriging. The basic kriging technique uses a weighted average of neighbouring samples to estimate the unknown value at a given location. It estimates weights based on the variogram. And the quality of the estimated surface is reflected in the quality of the weights. More specifically, you want weights that give an unbiased prediction and the smallest variance.

Unlike kriging that fits one whole model for an entire data set, Empirical Bayesian Kriging (EBK) simulates at least one hundred local models by sub-setting the whole data set. Because the model can morph itself locally to fit each individual semi-variogram using kriging methodology, it overcomes the challenge of stationarity. Each semi-variogram varies from each other. In the end, it mixes all the semi-variograms for a final surface. You can see the trends in the resulting data and use that to justify your selection.

Image segmentation and clustering with K-means

By far, the K-means clustering algorithm is one of the most popular methods of clustering data. This is an unsupervised algorithm and is used to segment the interest area from the unlabelled data into the number of groups represented by the variable K. But before applying the K-means algorithm, the first partial stretching enhancement is applied to the image to improve its quality. The algorithm iteratively assigns each data point into one of the K groupings, based on a similarity of features. For example, similarity can be based on spectral characteristics and location. In an unsupervised classification, the